

## Research Interests

**Socially Responsible Operations, Fairness, Pricing, and Urban Systems**

## Education

- Aug. 2021 – **Ph.D. in Operations Research**, *Columbia University*, New York, USA  
present Advisor: Prof. Adam N. Elmachtoub
- Sept. 2018 – **M.S. in Industrial & Systems Engineering**, *KAIST*, Daejeon, Korea  
Aug. 2020 Advisor: Prof. Il-Chul Moon
- Feb. 2012 – **B.S. in Industrial & Systems Engineering and Electrical Engineering**, *KAIST*, Daejeon, Korea  
Aug. 2018 Summa Cum Laude, Graduated first in the Department of Industrial & Systems Engineering

## Publications

- [1] **Fair Aggregation in Virtual Power Plants**  
Liudong Chen\*, Hyemi Kim\*, Adam N. Elmachtoub, Bolun Xu  
(Non-Archival) *Conference on Fairness, Accountability, and Transparency (FAccT)*, 2026.  
*Submitted to Management Science*
- [2] **Learning Fair Demand Models**  
Adam N. Elmachtoub, Hyemi Kim, Jonathan Y. Tan ( $\alpha$ - $\beta$ )  
(Non-Archival) *Conference on Fairness, Accountability, and Transparency (FAccT)*, 2026. *Submitted to Operations Research*
- [3] **Fair Fares for Vehicle Sharing Systems**  
Adam N. Elmachtoub and Hyemi Kim ( $\alpha$ - $\beta$ )  
*Operations Research*, 2026.  
(Non-Archival) *Conference on Fairness, Accountability, and Transparency (FAccT)*, 2025.  
○ Finalist, *INFORMS DEI Best Student Paper Award*, 2024  
○ Finalist, *INFORMS Transportation Science and Logistics Society Best Student Paper Award*, 2024
- [4] **Black-Box EM Algorithm for Estimating Latent States of High-Speed Vehicles**  
Yoon-Yeong Kim, Hyemi Kim, WonSung Lee, Han-Lim Choi, and Il-Chul Moon  
*American Institute of Aeronautics and Astronautics*, 2021.
- [5] **Counterfactual Fairness with Disentangled Causal Effect Variational Autoencoder**  
Hyemi Kim, Seungjae Shin, JoonHo Jang, Kyungwoo Song, Weonyoung Joo, Wanmo Kang, and Il-Chul Moon  
*Association for the Advancement of Artificial Intelligence (AAAI)*, 2021.
- [6] **Neutralizing Gender Bias in Word Embedding with Latent Disentanglement and Counterfactual Generation**  
Seungjae Shin, Kyungwoo Song, Joonho Jang, Hyemi Kim, Weonyoung Joo, and Il-Chul Moon  
*Findings of Empirical Methods in Natural Language Processing (EMNLP)*, 2020.
- [7] **Deep Generative Positive-Unlabeled Learning under Selection Bias**  
ByeongHu Na, Hyemi Kim, Kyungwoo Song, Weonyoung Joo, Yoonyeong Kim, and Il-Chul Moon  
*Conference on Information and Knowledge Management (CIKM)*, 2020.

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## Teaching Assistantships

| Year         | Course, School (Mean overall quality rating out of 5.0, if available) |
|--------------|-----------------------------------------------------------------------|
| Fall 2024    | Introduction to Probability and Statistics, Columbia IEOR (4.67)      |
| Spring 2024  | Stochastic Models, Columbia IEOR (4.64)                               |
| Fall 2023    | Analytics in Action, Columbia Business School                         |
| Spring 2023  | Transportation Analytics & Logistics, Columbia IEOR (4.6)             |
| Spring 2022  | Causal Inference, Columbia IEOR (4.4)                                 |
| Fall 2018/19 | Data Structure, KAIST ISysE                                           |

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## Honors and Awards

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|------|--------------------------------------------------------------------------------------------------------------------------------|
| 2025 | Columbia-Dream Sports AI Innovation Fellowship, Columbia University                                                            |
| 2024 | Finalist, INFORMS DEI Best Student Paper Award                                                                                 |
| 2024 | Finalist, INFORMS Transportation Science and Logistics Society Best Student Paper Award                                        |
| 2023 | Finalist, INFORMS 2023 BSS Data Challenge Competition (Report: Decoupling Price and Demand Estimation for Profit Optimization) |
| 2023 | Deming Doctoral Fellowship, Columbia University                                                                                |
| 2019 | Songhyun Award, KAIST                                                                                                          |
| 2012 | Presidential Science Scholarship, KFAS                                                                                         |

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## Academic Service

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|-------------|------------------------------------------------------------------------------------------|
| 2026        | Session Chair, INFORMS Annual Meeting                                                    |
| 2026        | Session Chair, Fairness Accountability and Transparency (FAccT)                          |
| 2026        | Reviewer, Fairness Accountability and Transparency (FAccT)                               |
| 2025 – 2026 | Program Committee, IJCAI – Special Track on AI and Social Good                           |
| 2024 – 2026 | Program Committee, Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO) |
|             | Reviewer, Socio-Economic Planning Sciences                                               |
|             | Reviewer, Journal of Risk and Insurance                                                  |
| 2014        | President, Student Council, Department of Industrial Engineering, KAIST                  |

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## Industry Experience

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|-------------|---------------------------------------------------------------------------------|
| Jun. 2024 – | <b>Data Scientist Intern</b> , <i>MOLOCO, Inc.</i> , Redwood City, USA          |
| Aug. 2024   | Designed and executed budget pacing simulations for online advertisements.      |
| Jan. 2021 – | <b>AI Research Intern</b> , <i>NAVER Corporation</i> , Seongnam, Korea          |
| Jul. 2021   | Studied test-time training methods for robust machine learning.                 |
| Jun. 2015 – | <b>Intern</b> , <i>Samsung Electronics</i> , Suwon, Korea                       |
| Aug. 2015   | Analyzed smart car development trends and improved patient analysis algorithms. |

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## Talks

### Fair Aggregation in Virtual Power Plants

- 2026 INFORMS RMP Conference, Ann Arbor, Michigan
- 2026 ACM FAccT Conference, Montreal, Canada
- 2026 Market Innovation Workshop, Virtual

### Learning Fair Demand Models

- 2025 INFORMS Annual Meeting, Atlanta, Georgia
- 2025 INFORMS RMP Conference, New York, NY

### Fair Fares for Vehicle Sharing Systems

- 2025 ACM FAccT Conference, Athena, Greek
- 2025 Seminar at KAIST, Daejeon, Korea
- 2025 INFORMS TSL Workshop, Seoul, Korea
- 2024 INFORMS DEI Best Student Paper Awards, Seattle, WA
- 2024 INFORMS TSL Best Student Paper Awards, Seattle, WA
- 2024 INFORMS RMP Conference, Los Angeles, CA
- 2024 Market Innovation Workshop, Virtual
- 2023 INFORMS Annual Meeting, Phoenix, AZ
- 2023 Workshop on AI & Analytics for Social Good, College Park, MD

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## Skills & Personal

- Programming Python, MATLAB, C; PyTorch, TensorFlow, Gurobi, Pyomo
- Languages English (advanced), Korean (native)
- Hobby Brazilian jiu-jitsu (purple belt)